

HDOC - Day 10

Time Conversion and Leap Year

Question 1 - Convert Seconds into Days, Hours, Minutes and Seconds

Problem understanding: Input total time in seconds. Convert it into complete days, remaining hours, remaining minutes and seconds.

Formula: days = total/86400; remainder = total%86400; hours = remainder/3600; minutes = remaining/60; seconds = remaining%60.

Algorithm

1. Start. 2. Read total seconds. 3. If seconds is negative, display an error and stop. 4. Calculate days by dividing by 86400. 5. Find the remainder. 6. Calculate hours from the remainder using 3600. 7. Calculate minutes using 60. 8. The final remainder gives seconds. 9. Display days, hours, minutes and seconds. 10. Stop.

Pseudocode

```
READ total_seconds
IF total_seconds < 0 THEN print "Invalid input" and STOP
days = total_seconds / 86400
rem = total_seconds % 86400
hours = rem / 3600
rem = rem % 3600
minutes = rem / 60
seconds = rem % 60
PRINT days, hours, minutes, seconds
```

Flowchart

START -> Input total seconds -> Negative? -> Yes: Error -> STOP; No -> Calculate days -> hours -> minutes -> seconds -> Display converted time -> STOP.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	90061 s	1 d, 1 h, 1 min, 1 s	1 d, 1 h, 1 min, 1 s	Pass
2	3665 s	0 d, 1 h, 1 min, 5 s	0 d, 1 h, 1 min, 5 s	Pass
3	60 s	0 d, 0 h, 1 min, 0 s	0 d, 0 h, 1 min, 0 s	Pass

C Program

```
#include <stdio.h>
int main() {
    long total,days,hours,minutes,seconds,rem;
    printf("Enter time in seconds: ");
    scanf("%ld",&total);
    if(total<0) { printf("Invalid input.\n"); return 0; }
    days=total/86400;
    rem=total%86400;
    hours=rem/3600;
    rem=rem%3600;
    minutes=rem/60;
    seconds=rem%60;
    printf("Days      : %ld\n",days);
    printf("Hours      : %ld\n",hours);
    printf("Minutes    : %ld\n",minutes);
    printf("Seconds    : %ld\n",seconds);
    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang time.c -o time
puppie@Anvikas-MacBook-Air Desktop % ./time
Enter time in seconds: 90061
Days      : 1
Hours      : 1
Minutes    : 1
Seconds    : 1
```

```
puppie@Anvikas-MacBook-Air Desktop % ./time
Enter time in seconds: 3665
Days      : 0
Hours      : 1
Minutes    : 1
Seconds    : 5
```

```
puppie@Anvikas-MacBook-Air Desktop % ./time
Enter time in seconds: 60
Days      : 0
Hours     : 0
Minutes   : 1
Seconds   : 0
```

Result: The program converted time in seconds into days, hours, minutes and seconds correctly for all prepared test cases.

Question 2 - Check Whether a Year is a Leap Year

Problem understanding: Input a year and determine whether it is a leap year. A year divisible by 400 is a leap year. Otherwise, it is a leap year only when divisible by 4 but not by 100.

Algorithm

1. Start. 2. Read year. 3. If year is divisible by 400, it is a leap year. 4. Otherwise, if it is divisible by 4 and not divisible by 100, it is a leap year. 5. Otherwise it is not a leap year. 6. Display the result. 7. Stop.

Pseudocode

```
READ year
IF year % 400 == 0
    PRINT "Leap year"
ELSE IF year % 4 == 0 AND year % 100 != 0
    PRINT "Leap year"
ELSE
    PRINT "Not a leap year"
```

Flowchart

START -> Input year -> Divisible by 400? -> Yes: Leap year -> STOP; No -> Divisible by 4 AND not by 100? -> Yes: Leap year -> No: Not leap year -> STOP.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	2024	Leap year	Leap year	Pass
2	1900	Not a leap year	Not a leap year	Pass
3	2000	Leap year	Leap year	Pass

C Program

```
#include <stdio.h>
int main() {
    int year;
    printf("Enter a year: ");
    scanf("%d",&year);
    if(year%400==0)
        printf("%d is a leap year.\n",year);
    else if(year%4==0 && year%100!=0)
        printf("%d is a leap year.\n",year);
    else
        printf("%d is not a leap year.\n",year);
    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang leap.c -o leap
puppie@Anvikas-MacBook-Air Desktop % ./leap
Enter a year: 2024
2024 is a leap year.
```

```
puppie@Anvikas-MacBook-Air Desktop % ./leap
Enter a year: 1900
1900 is not a leap year.
```

```
puppie@Anvikas-MacBook-Air Desktop % ./leap
Enter a year: 2000
2000 is a leap year.
```

Result: The leap-year program was compiled, executed and tested successfully against the prepared cases.